

Estimation for $ARMA(p,q)$ models^{[1],[2]}

We have n observations, $x_1, \dots, x_n$, from a causal and invertible $ARMA(p, q)$ process, where p and q are known.

We want to estimate the parameters of the model $\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q$ and σ_w^2 .

[1] Robert Shumway, David Stoffer. Time Series Analysis and its Applications with R Examples, 4th edition, Springer, 2017 (chapter 3)
[2] Peter Brockwell, Richard Davis. Time Series: Theory and Methods, Springer-Verlag, 1987 (chapter 8)

I. Method of moments

It consists of replacing population moments by sample moments and then determining the parameters in term of the sample moments.

$$E(X) \text{ replaced by } \bar{x} = \frac{1}{n} \sum_{t=1}^n x_t - \text{sample mean}$$

$$\gamma(h) \text{ replaced by } \hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x})(x_t - \bar{x})$$

$$(good \text{ for } n \geq 50, h \leq \frac{n}{4})$$

I. Method of moments

For an AR(p) model, recall that

$$\left. \begin{aligned} \gamma(h) &= \emptyset_1 \gamma(h-1) + \dots + \emptyset_p \gamma(h-p), \quad h = 1, \dots, p \\ \sigma_w^2 &= \gamma(0) - \emptyset_1 \gamma(1) - \dots - \emptyset_p \gamma(p) \end{aligned} \right\} \text{Yule-Walker equations}$$

In matrix form, the Yule-Walker equations are:

$$\Gamma_p \emptyset = \gamma_p, \quad \sigma_w^2 = \gamma(0) - \emptyset' \gamma_p, \text{ where}$$

$$\Gamma_p = \begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(p-1) \\ \vdots & & & \vdots \\ \gamma(p-1) & \dots & \dots & \gamma(0) \end{pmatrix} \quad \emptyset = \begin{pmatrix} \emptyset_1 \\ \vdots \\ \emptyset_p \end{pmatrix} \quad \gamma_p = \begin{pmatrix} \gamma(1) \\ \vdots \\ \gamma(p) \end{pmatrix}$$

We replace $\gamma(h)$ by $\hat{\gamma}(h)$.

I. Method of moments

$$\left. \begin{aligned}\hat{\varnothing} &= \hat{\Gamma}_p^{-1} \cdot \hat{\gamma}_p \\ \hat{\sigma}_w^2 &= \hat{\gamma}(0) - \hat{\gamma}_p' \hat{\Gamma}_p^{-1} \hat{\gamma}_p\end{aligned}\right\} \text{the Yule-Walker estimators}$$

or, equivalent

$$\begin{aligned}\hat{\varnothing} &= \hat{R}_p^{-1} \hat{\rho}_p \\ \hat{\sigma}_w^2 &= \hat{\gamma}(0)[1 - \hat{\rho}_p' \hat{R}_p^{-1} \hat{\rho}_p],\end{aligned}$$

where $\hat{R}_p = \frac{\hat{\Gamma}_p}{\hat{\gamma}(0)}$ and $\hat{\rho}_p = \frac{\hat{\gamma}_p}{\hat{\gamma}(0)}$.

I. Method of moments

Proposition - For a causal AR process, the asymptotic behavior of the Yule-Walker estimators is normal:

$$\begin{aligned}\sqrt{n}(\hat{\theta} - \theta) &\xrightarrow{n \rightarrow \infty} N(0, \sigma_w^2 \Gamma_p^{-1}) \\ \hat{\sigma}_w^2 &\xrightarrow{n \rightarrow \infty} \sigma_w^2\end{aligned}\quad (n \text{ is the sample size})$$

We can avoid calculating the inverse of $\hat{\Gamma}_p$ (or $\hat{R}_p$) by replacing $\gamma(h)$ with $\hat{\gamma}(h)$ in the Durbin-Levinson algorithm.

I. Method of moments

(from C4) The Durbin-Levison algorithm

$$\phi_{11} = \frac{\gamma(1)}{\gamma(0)}$$

$$V_0 = \gamma(0)$$

$$\phi_{nn} = \frac{\gamma(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \gamma(n-k)}{V_{n-1}}, n \geq 1$$

$$\phi_{nk} = \phi_{n-1,k} - \phi_{nn} \phi_{n-1,n-k}, \quad k = 1, \dots, n-1, \quad n \geq 2$$

$$V_n = V_{n-1} [1 - \phi_{nn}^2]$$

(here n is the iteration)

Order of computation:

$$\phi_{11}, V_1, \phi_{22}, \phi_{21}, V_2, \phi_{33}, \phi_{31}, \phi_{32}, V_3 \dots$$

The predictor: $\hat{X}_{n+1} = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}X_1$

I. Method of moments

Proposition - For a causal AR(p) process, we have

$$\sqrt{n} \cdot \hat{\varnothing}_{hh} \xrightarrow{n \rightarrow \infty} N(0,1) \text{ for all } h > p. \quad (n \text{ is the sample size})$$

Obs. The Yule-Walker estimators, computed iteratively in the Durbin-Levinson algorithm, are:

$$\begin{aligned} \hat{\varnothing}_p &= (\hat{\varnothing}_{p1}, \dots, \hat{\varnothing}_{pp}) \\ \sqrt{n}(\hat{\varnothing}_p - \varnothing) &\xrightarrow{n \rightarrow \infty} N(0, \sigma_w^2 \Gamma_p^{-1}) \end{aligned}$$

We say that the estimator $\hat{\varnothing}_p$ is consistent for $\varnothing$.

I. Method of moments

The estimates $\hat{\varnothing}_{hh}$ are extremely valuable, for two reasons:

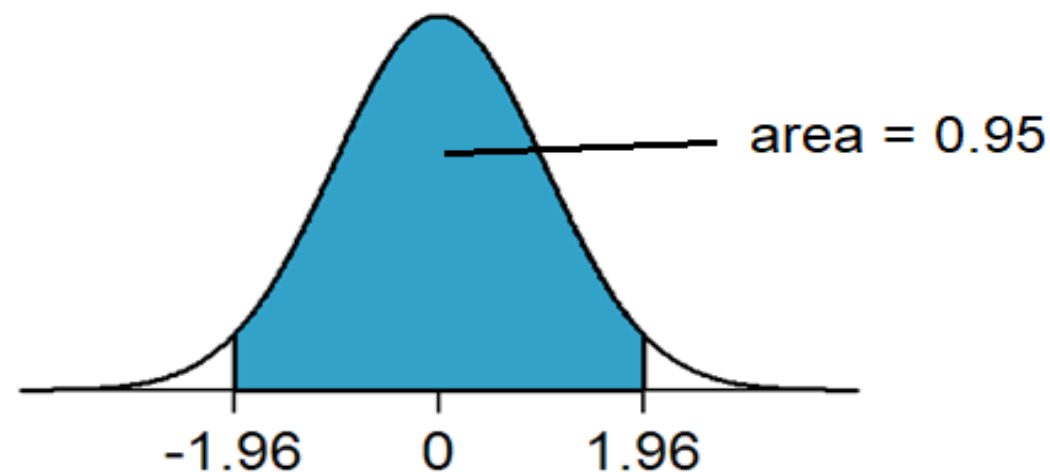
- to decide on the appropriateness of an AR model;
- then, to choose an appropriate order for the model to be fitted.

For an AR(p) process, the partial autocorrelations $\varnothing_{mm} = 0$ for $m > p$.

The estimator $\hat{\varnothing}_{mm} \xrightarrow{n \rightarrow \infty} N\left(0, \frac{1}{n}\right)$ for $m > p$.

I. Method of moments

If $X \sim N(\mu, \sigma^2)$ then $-1.96 < \frac{x-\mu}{\sigma} < 1.96$ with probability 0.95



This suggests a way to choose a preliminary value of p :

Set $p = P$ (a large value, for example, $P=100$) initially; n is the sample size

repeat

if $|\hat{\theta}_{pp}| < 1.96 \cdot n^{-\frac{1}{2}}$ then it means that $\theta_{pp} = 0$

$p = p - 1$

until $|\hat{\theta}_{pp}| \geq 1.96 \cdot n^{-\frac{1}{2}}$

I. Method of moments

The Yule-Walker estimators for $AR(p)$ models are optimal as $AR(p)$ models are linear models and Y-W estimators are essentially least squares estimators.

```
x=arima.sim(list(order=c(2,0,0), ar=c(1,-.9)), n=500)
```

```
x.yw = ar.yw(x, order=2)
```

```
x.yw$ar #coefficient estimates
```

```
[1] 0.9921305 -0.9003378
```

```
x.yw$var.pred #wn variance estimate
```

```
[1] 1.076749
```

I. Method of moments

The method of moments for MA or ARMA models will not give optimal estimator because these models are nonlinear in the parameters.

- for an invertible MA(1) model:

$$X_t = \sum_{j=1}^{\infty} (-\theta)^j X_{t-j} + w_t \quad - \text{ is nonlinear in } \theta$$

I. Method of moments

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Obs. Analogically to the estimation for AR models using the Durbin-Levinson algorithm, the innovation algorithm can be used to estimate the parameters of an MA(q)

$$\hat{\theta}_q = (\hat{\theta}_{q1}, \dots, \hat{\theta}_{qq})' \text{ estimator for } \theta_q = (\theta_1, \dots, \theta_q)'$$

But there is one important difference – for MA(q), the estimator $\hat{\theta}_q$ is not consistent for the true parameter vector θ_q .

II. Maximum likelihood and Least squares estimations for ARMA models (MLE and LSE)

Let $\mathbf{X}_n = (X_1, \dots, X_n)'$

$$\hat{\mathbf{X}}_n = (\hat{X}_1, \dots, \hat{X}_n)',$$

where $\hat{X}_1 = 0$ and

$$\hat{X}_j = P_{\overline{sp}\{X_1, \dots, X_{j-1}\}} X_j, \quad j \geq 2$$

$\Gamma_n = E(\mathbf{X}_n \cdot \mathbf{X}_n')$ – assume that is non-singular

If $\{\mathbf{X}_t\}$ is Gaussian (i.e., the distribution functions are all multivariate normal) with mean zero, the likelihood of $\mathbf{X}_n$ is:

$$L(\Gamma_n) = (2\pi)^{-\frac{n}{2}} \cdot (\det \Gamma_n)^{-\frac{1}{2}} \cdot \exp\left(-\frac{1}{2} \mathbf{X}_n' \Gamma_n^{-1} \mathbf{X}_n\right)$$

II. MLE and LSE

If $\{X_t\}$ is not Gaussian, $L(\Gamma_n)$ is still a measure of the goodness of fit of the covariance matrix Γ_n to the data.

$$\Gamma_n = \Gamma_n(\phi, \theta, \sigma_w^2)$$

$$\phi = (\phi_1, \dots, \phi_p)'$$

$$\theta = (\theta_1, \dots, \theta_q)'$$

We want to maximize $L(\Gamma_n)$ with respect to $(\phi, \theta, \sigma_w^2)$.

II. MLE and LSE

The direct calculation of $\det \Gamma_n$ and Γ_n^{-1} in $L(\Gamma_n)$ can be avoided, using instead the recursive formulas from the innovation algorithm.

The mean squared error $E \left[(X_{n+1} - \hat{X}_{n+1})^2 \right] = v_n$

$$r_n \stackrel{\text{def}}{=} \frac{v_n}{\sigma_w^2}$$

II. MLE and LSE

The likelihood function can be written as (for details, see Brockwell & Davis, pages 247-250):

$$\begin{aligned} L(\Gamma_n) &= L(\emptyset, \theta, \sigma_w^2) \\ &= (2\pi\sigma_w^2)^{-\frac{n}{2}} \cdot (r_0 \cdot r_1 \cdot \dots \cdot r_{n-1})^{-\frac{1}{2}} \cdot \exp \left(-\frac{1}{2} \sigma_w^{-2} \sum_{j=1}^n (X_j - \hat{X}_j)^2 / r_{j-1} \right) \end{aligned}$$

\\ not
 $S(\emptyset, \theta)$

$$\ln L(\emptyset, \theta, \sigma_w^2) = -\frac{n}{2} \ln(2\pi\sigma_w^2) - \frac{1}{2} \sum_{j=1}^n \ln r_{j-1} - \frac{1}{2} \sigma_w^{-2} S(\emptyset, \theta)$$

$\hat{X}_j$ and r_j are independent of σ_w^2

II. MLE and LSE

$$\frac{\partial \ln L(\emptyset, \theta, \sigma_w^2)}{\partial \sigma_w^2} = 0$$

$$-\frac{n}{2} \cdot \frac{1}{\sigma_w^2} + \frac{1}{2} \cdot \frac{1}{(\sigma_w^2)^2} \cdot S(\emptyset, \theta) = 0$$

$$\hat{\sigma}_w^2 = \frac{1}{n} \cdot S(\hat{\emptyset}, \hat{\theta}),$$

where $\hat{\emptyset}$ and $\hat{\theta}$ are the values of $\emptyset, \theta$ which minimize

$$l(\emptyset, \theta) = \ln(n^{-1}S(\emptyset, \theta)) + n^{-1} \sum_{j=1}^n \ln r_{j-1}$$

$l(\emptyset, \theta)$ is called the concentrated (or reduced) likelihood.

$\hat{\emptyset}$ and $\hat{\theta}$ that minimize $l(\emptyset, \theta)$ are determined using non-linear numerical optimization algorithms (e.g., Newton-Raphson algorithm).

II. MLE and LSE

An alternative estimation procedure is to minimize $S(\emptyset, \theta)$ with respect to $\emptyset$ and θ . The estimators obtained will be referred to as the “least squares” estimators $\tilde{\emptyset}$ and $\tilde{\theta}$ of $\emptyset$ and θ .

Proposition – For causal and invertible ARMA processes, the maximum likelihood and the least squares estimators provide optimal estimators of σ_w^2 , $\emptyset$ and θ .

II. MLE and LSE

```
x=arima.sim(list(order=c(2,0,1), ar=c(1,-.9),ma=c(0.9)), n=100)
```

```
v=arima(x, order = c(2,0,1))
```

```
v$coef ar1 ar2 ma1 intercept  
1.0061145 -0.8754776 0.9847590 -0.3537682
```

```
v$sigma2  
[1] 0.8888034
```

```
x=arima.sim(list(order=c(2,0,1), ar=c(1,-.9),ma=c(1.7)), n=100)
```

```
v=arima(x, order = c(2,0,1))
```

```
v$coef ar1 ar2 ma1 intercept  
0.9809513 -0.8705035 0.6825593 0.4325465
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[1] 2.80407
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causal,
invertible

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```

causal,
non-invertible

Dealing with non-stationary time series

Two approaches were developed in order to make a non-stationary time series stationary (Brockwell and Davis, pages 14-25):

- estimate and extract the deterministic components m_t and s_t , with the hope that the residual y_t will be stationary:

$$x_t = m_t + s_t + y_t$$

- apply difference operators repeatedly to the data x_t until we get a realization of a stationary process $\nabla x_t = x_t - x_{t-1}$

Dealing with non-stationary time series

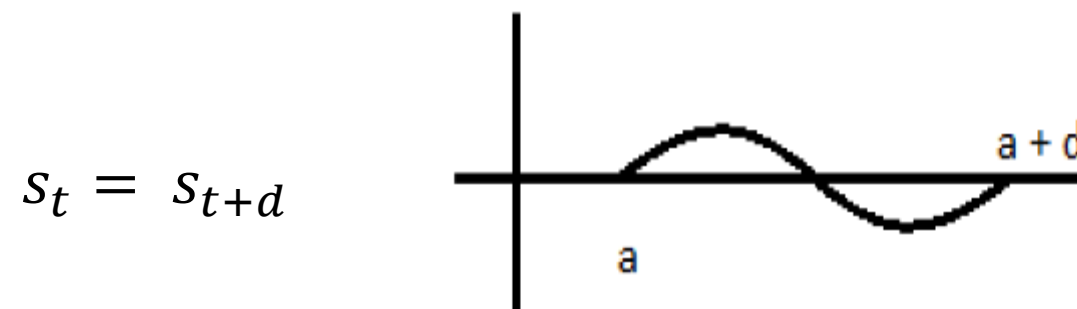
We assume that the non stationary $\{x_t\}$ is of the form

$$x_t = m_t + s_t + y_t,$$

where m_t is a slowly changing function called “trend component”

s_t is the “seasonal component” – is a function with known period d

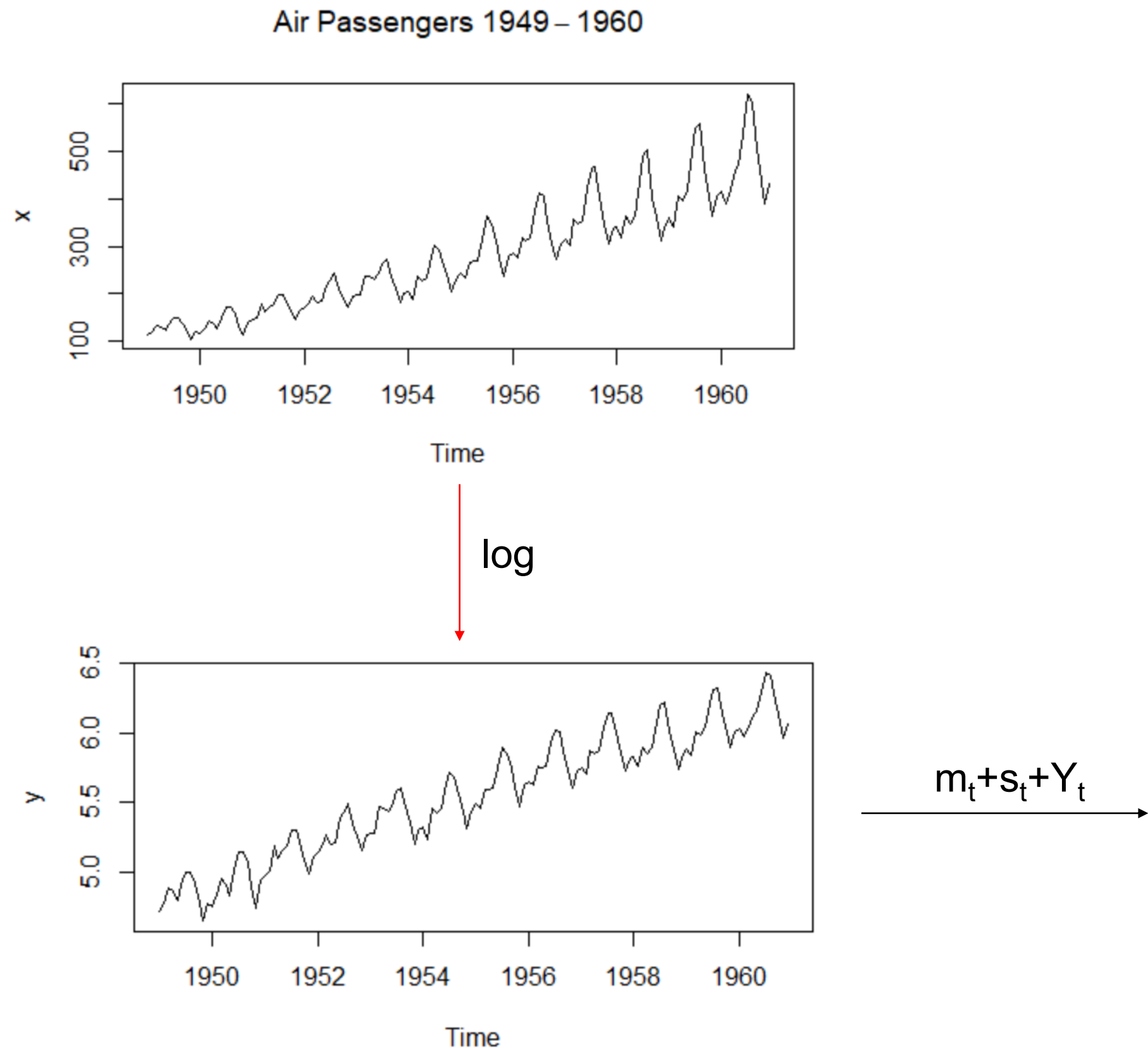
$$\text{and } \sum_{t=a}^{a+d} s_t = 0$$



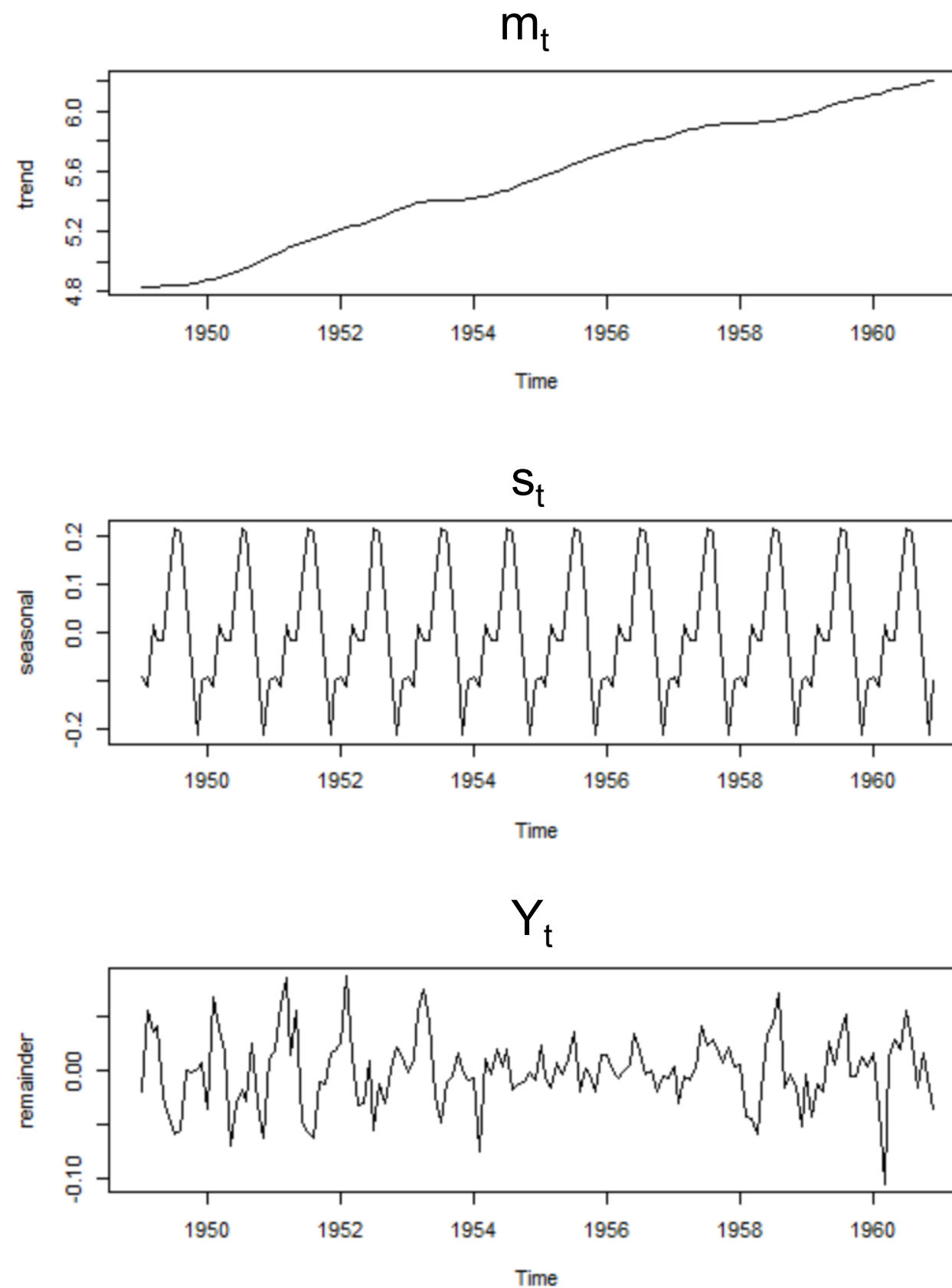
y_t is the random noise component – is stationary.

In practice, preliminary transformation of the data (e.g., log, square root, cube root etc.) may be necessary to make the transformed data compatible with this model.

Dealing with non-stationary time series



Dealing with non-stationary time series



Dealing with non-stationary time series

The ARIMA(p, d, q) model

It is a process which, after differencing a number of times, reduces to an ARMA(p, q) process.

Def. If d is a non-negative integer, then $\{X_t\}$ is said to be an ARIMA(p, d, q) process if $Y_t = (1 - B)^d X_t$ is a causal ARMA(p, q) process.

$$\phi(B)(1 - B)^d X_t = \theta(B)w_t, \quad w_t \sim \text{wn}(0, \sigma_w^2)$$

Dealing with non-stationary time series

Example $\{X_t\}$ is an ARIMA(1,1,0) process if

$$(1 - \phi B)(1 - B)X_t = w_t, \quad \phi \in (-1, 1)$$

$$(1 - B)X_t = Y_t$$

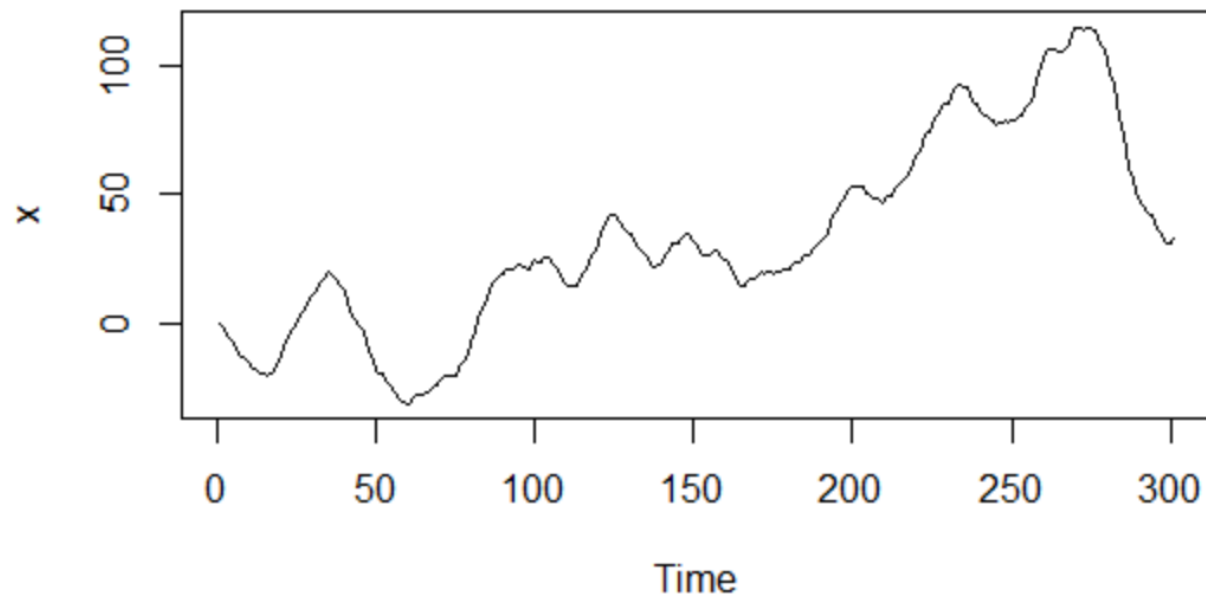
$$(1 - \phi B)Y_t = w_t \text{ is causal as } \phi \in (-1, 1)$$

Obs. Estimation of ϕ , θ and σ_w^2 is based on the observed differences $(1 - B)^d X_t$.

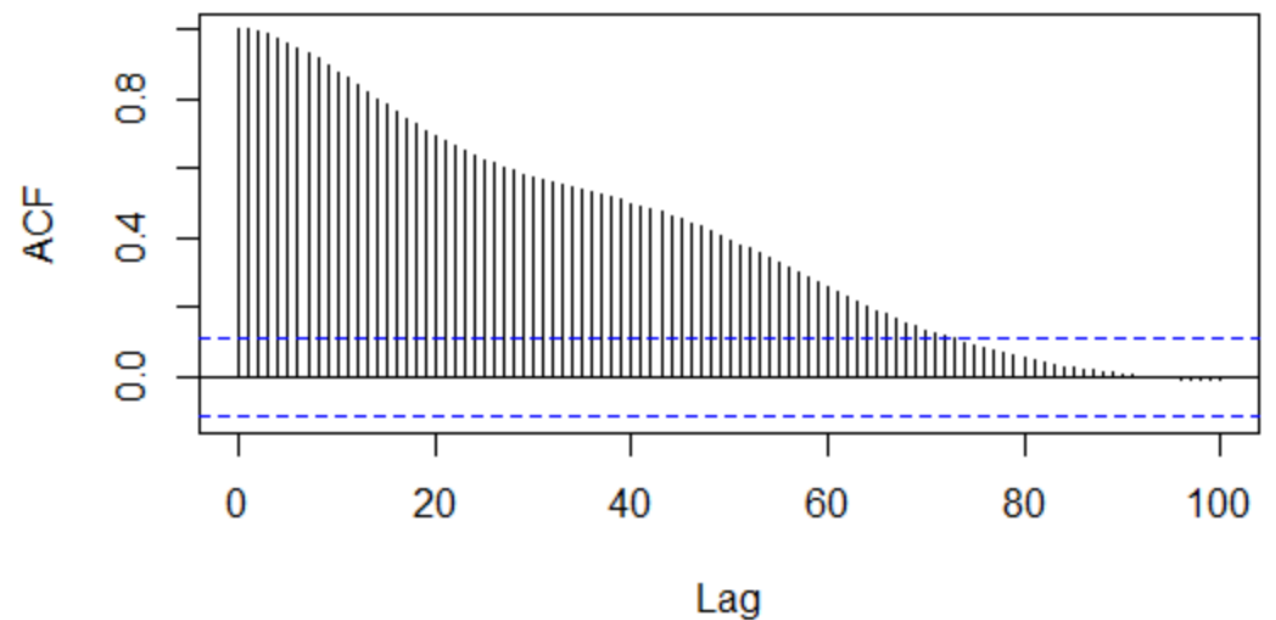
Dealing with non-stationary time series

Obs. A slowly decaying positive ACF is a distinctive feature of the data which suggests the appropriateness of an ARIMA model.

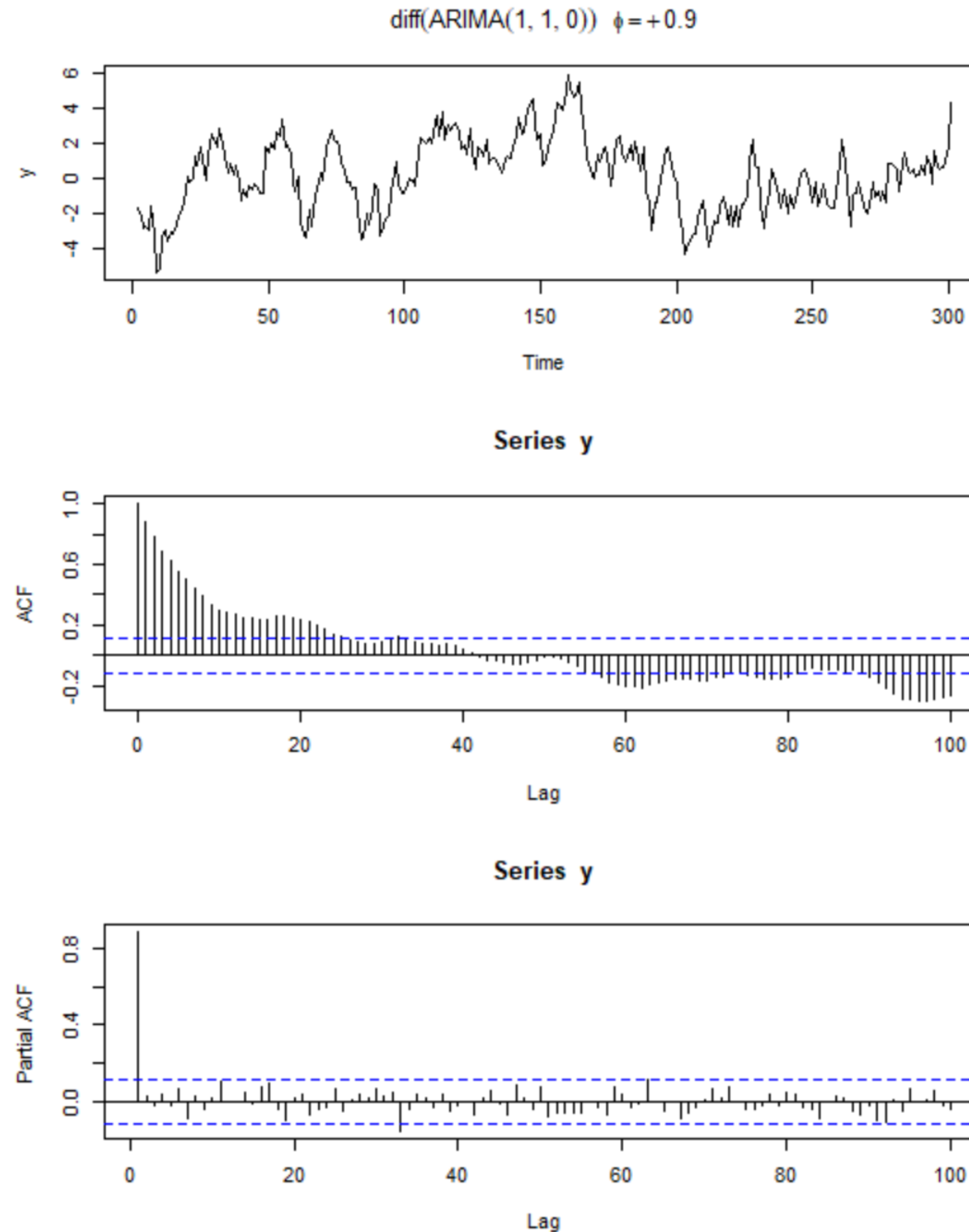
ARIMA(1, 1, 0) $\phi = +0.9$



Series x



Dealing with non-stationary time series



Dealing with non-stationary time series

1) Random Walk with Drift is not stationary

$$X_t = \delta + X_{t-1} + w_t, \quad w_t \sim \text{wn}(0, \sigma_w^2)$$

Differencing will lead to a stationary process

$$\nabla X_t = X_t - X_{t-1} = \delta + w_t$$

Dealing with non-stationary time series

$$2) X_t = m_t + Y_t$$

$$m_t = \beta_0 + \beta_1 t$$

Y_t is stationary

$$\nabla X_t = \beta_1 + Y_t - Y_{t-1} = \beta_1 + \nabla Y_t \text{ is stationary}$$

Examples of (non) stationary time series*

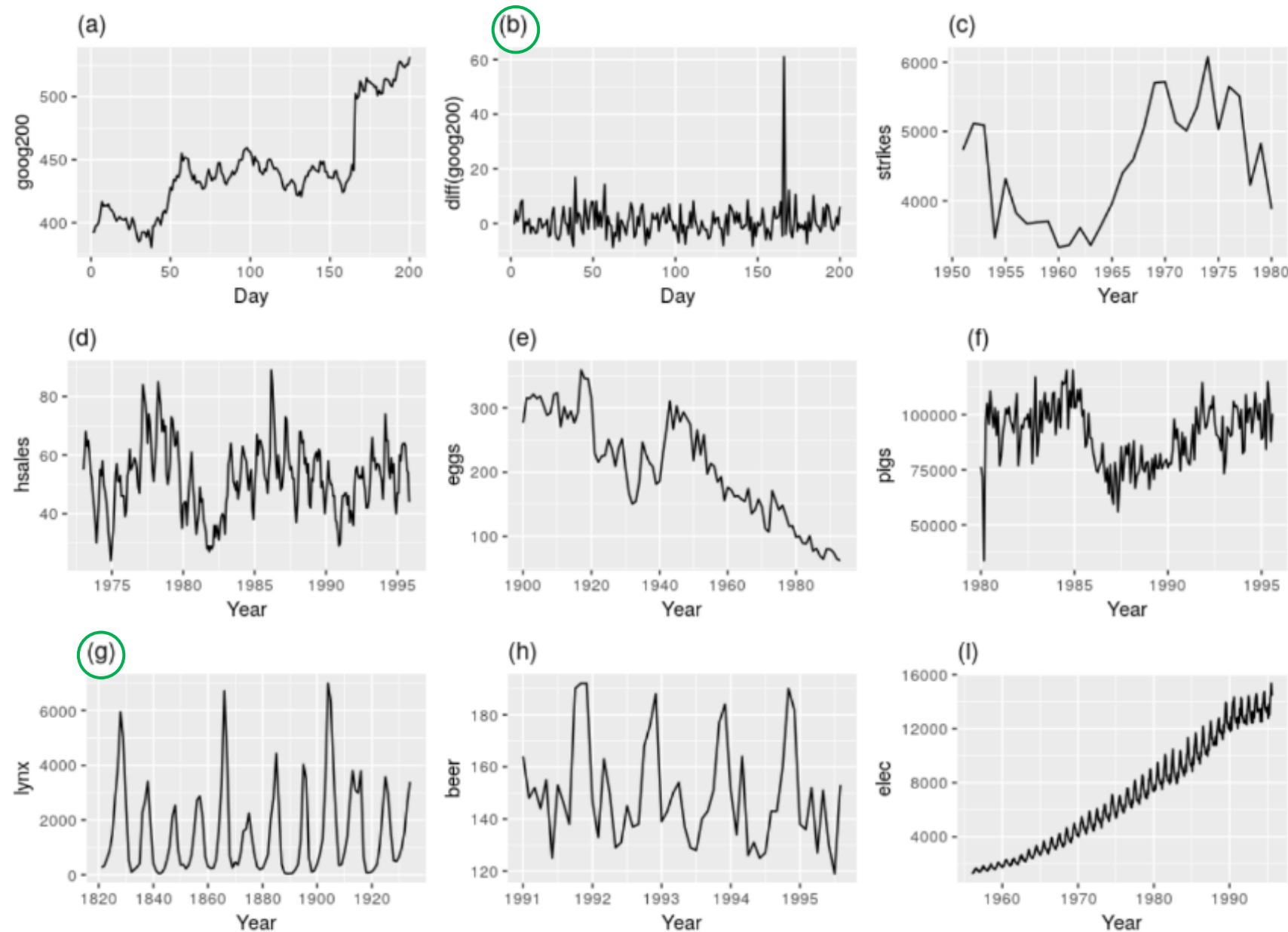


Figure 1: Nine examples of time series data; (a) Google stock price for 200 consecutive days; (b) Daily change in the Google stock price for 200 consecutive days; (c) Annual number of strikes in the US; (d) Monthly sales of new one-family houses sold in the US; (e) Annual price of a dozen eggs in the US (constant dollars); (f) Monthly total of pigs slaughtered in Victoria, Australia; (g) Annual total of lynx trapped in the McKenzie River district of north-west Canada; (h) Monthly Australian beer production; (i) Monthly Australian electricity production. [Hyndman & Athanasopoulos, 2018]

*taken from <https://towardsdatascience.com/detecting-stationarity-in-time-series-data-d29e0a21e638>